Atanu Adhikari

AI/ML & Full-Stack Developer · Seeking a remote AI/ML internship

Rishihood University, Sonipat, Haryana, India (IST) · atanuadhikari5@gmail.com · atanu.online · github.com/techatanu · LinkedIn: atanu-adhikari-3298492ab

SUMMARY

B.Tech Computer Science & AI student (3rd year) who builds end-to-end AI/ML projects, from data and models to deployed apps. Experience with classical ML (scikit-learn), deep learning (PyTorch), LLM agents (LangGraph, RAG, ChromaDB) and full-stack web apps (FastAPI, React, Node.js). Reports honest, reproducible metrics.

PROJECTS

Intelligent Exam Question Analyser — NLP · Agentic RAG · Team of 4 · Apr 2026

- Built the dataset (6,200 rows → 1,996 unique) and TF-IDF features; handled imbalance with SMOTE after the split to avoid leakage. Logistic Regression reached **71% macro accuracy** vs XGBoost 54.5% and Random Forest 51.8%.
- Built a 3-node LangGraph agent (retrieve → analyse → format) over a ChromaDB knowledge base (MiniLM embeddings) with Llama 3.3 via Groq, producing grounded five-section reports. Live on Streamlit.

Infant State Recognition from Cry Audio — Audio ML · Python, librosa, scikit-learn, PyTorch

- Classified cries into 8 need states using 218 librosa features and a cost-sensitive Random Forest (test macro F1 **38.1%**, accuracy 78.6%) on 511 clips with a 54:1 class imbalance.
- Benchmarked a PyTorch CNN on log-mel spectrograms with SpecAugment and audio augmentation (macro F1 25.8%); found and quantified two data leaks via grouped cross-validation; added pytest + GitHub Actions CI and a Gradio demo.

Student GitHub Reviewer — LangGraph, Llama 3.1 (Groq), FastAPI, Streamlit

- AI agent that reviews a student's GitHub portfolio; LangGraph pipeline served through a FastAPI backend on Render with a Streamlit UI.

Cognify: AI Career Guidance — Construct Hackathon · Dec 2025

- Co-built an AI agent that asks cognitive questions and suggests careers, roadmaps and colleges; launched on Product Hunt.

MediTrack & QuickCart — React, Node.js, Express, MongoDB, JWT

- MediTrack: health-record manager with JWT auth, role-based access and CRUD REST APIs (Vercel + Render). QuickCart: React cart with API data, real-time pricing and routing.

EXPERIENCE & LEADERSHIP

AI/ML & Full-Stack Developer (Project-Based) — Nov 2025 – Present

Self-directed projects · Remote: ML models, LLM apps and AI agents, plus MERN apps deployed end to end.

Acting Executive, DevClub — Newton School of Technology × Rishihood University

EDUCATION

Newton School of Technology — Aug 2024 – Aug 2028

B.Tech, Computer Science and Artificial Intelligence

SKILLS

ML / DL	Python, NumPy, Pandas, scikit-learn, PyTorch, XGBoost, SMOTE, librosa, NLP
LLMs & agents	LangGraph, LangChain, RAG, ChromaDB, sentence-transformers, Llama 3.x (Groq)
Backend / frontend	FastAPI, Node.js, Express, MongoDB, JWT, REST, React, Streamlit, Gradio
Tools	Git, GitHub Actions, pytest, Vercel, Render, nginx, Tableau, Jupyter
Certification	Introduction to Generative AI
Languages	English, Hindi, Bengali